

Popularising Summary

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Master's thesis: an extremal problem on connectivity in networks (Erdős Problem 915).

Many of the networks we rely on are useful precisely because they offer backup routes. If one road is closed, traffic takes another. If one cable fails, a message still arrives along a different chain of links. Mathematicians study this with graphs, where dots stand for objects and lines for the connections between them. This thesis asks a simple-sounding question: how many connections can a network with a fixed number of points have, if no two of those points may be joined by too many separate, non-overlapping routes?

That question is a version of a problem posed decades ago by Béla Bollobás and Paul Erdős, listed today as Erdős Problem 915. In its original form it concerns ordinary networks and routes that may not share a connection, and for that case the answer has been known since a theorem of Mader. This thesis starts from that result and then changes the rules of the network one at a time: connections may be doubled up, directions may be added so that a link works only one way, or a single connection may join more than two points at once. With each change the pencil-and-paper proof becomes longer and more delicate, until checking it by hand stops being realistic.

At that point the thesis hands the work to a computer. One program describes every version of the network in the same way, measures exactly how many separate routes run between any two points, and proves the answer outright for small networks by checking every case. Where checking everything is hopeless, it searches for good examples instead, walking from one network to a slightly better one while keeping a short memory of its recent moves so it cannot simply undo them. When the search is pointed at cases we already understand, it rediscovers the very networks the earlier proofs singled out.

The interplay between proof and experiment then pays off in five places. For networks whose connections may each tie several points together at once, the thesis settles the two strictest limits on backup routes. Under the strictest one the answer is exact at every size. Under the next one the same formula is always an upper limit, and is reached exactly once the network has room to spare, which the thesis pins down precisely. For one-way networks it proves, with no assumption about what the best network looks like, that the largest possible number of links is a quarter of the square of the number of points, give or take an amount growing only in proportion to that number. That settles the shape of an answer whose leading term had been only a conjecture, and what remains unknown is a single numerical factor. For networks whose links may be doubled up, and whose routes must avoid sharing any intermediate point as well, the thesis poses a question that had not been separated out before. It shows that the answer is decided by the best densely connected piece of each size, and settles it exactly under the strictest limits. For one-way networks whose links may be doubled up, the thesis settles the exact answer completely. The idea is to find, inside any such network, a thin skeleton that alone remembers who can reach whom. Removing it costs every pair at least one backup route, so peeling off one skeleton for each unit of the limit accounts for every link and proves the count outright. And for one-way networks with many-point connections, the obvious two-layer design turns out to be the best there is for large networks.